

Review Questions at rear of Chapter 13

- 1. What are decision support systems, and what role do they play in the business environment?**
- 2. Explain how the main components of a DSS interact to form a system.**
- 3. What are the most relevant differences between operational and decision support data?**
- 4. What is a data warehouse, and what are its main characteristics?**
- 5. Give three examples of problems likely to be found when operational data are integrated into the data warehouse.**

Use the following scenario to answer questions 6 through 13.

While working as a database analyst for a national sales organization, you are asked to be part of its data warehouse project team.

- 6. Prepare a high-level summary of the main requirements to evaluate DBMS products for data warehousing.**
- 7. Your data warehousing project group is arguing about prototyping a data warehouse before its implementation. The project group members are especially concerned about the need to acquire some data warehousing skills before implementing the enterprise-wide data warehouse. What would you recommend? Explain your recommendations.**
- 8. Suppose you are selling the data warehouse idea to your users. How would you explain to them what multidimensional data analysis is and explain its advantages?**
- 9. The Data Warehousing project group has invited you to provide an OLAP overview before making a commitment. The group's members are particularly concerned about the OLAP client/server architecture requirements and how OLAP will fit the existing environment. Your job is to explain to them the main OLAP client/server components and architectures.**
- 10. One of your vendors recommends using an MDBMS. How would you explain this recommendation to your project leader?**
- 11. The project group is ready to make a final decision between ROLAP and MOLAP. What should be the basis for this decision? Why?**

- 12. The data warehouse project is in the design phase. Explain to your fellow designers how you would use a star schema in the design.**
- 13. Trace the evolution of DSS from its origins to today's advanced analytical tools. What major technologies influenced this evolution?**
- 14. What is OLAP, and what are its main characteristics?**
- 15. Explain ROLAP, and give the reasons you would recommend its use in the relational database environment.**
- 16. Explain the use of facts, dimensions, and attributes in the star schema.**
- 17. Explain multidimensional cubes and describe how the slice and dice technique fits into this model.**

- 18. In the star schema context, what are attribute hierarchies and aggregation levels and what is their purpose?**
- 19. Discuss the most common performance improvement techniques used in star schemas.**
- 20. Explain some of the most important issues in data warehouse implementation.**
- 21. What is data mining, and how does it differ from traditional DSS tools?**
- 22. How does data mining work? Discuss the different phases in the data mining process.**